

## 32.4 Maximum likelihood and REML

Everything so far has treated the variance components as known. They never are. The moment estimates of [Section 32.1](#) work in balanced layouts and lose their footing in unbalanced ones, where no canonical set of quadratic forms presents itself. Under normality the likelihood has no such difficulty, but it has one defect, which this section diagnoses and repairs: maximum likelihood underestimates variance components, because it does not pay for the fixed effects it has fitted.

### 32.4.1 The likelihood and its profile

Under the normal linear mixed model,  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{V})$  with  $\mathbf{V} = \mathbf{V}(\boldsymbol{\theta})$ , so by Theorem 3.2.2

$$\ell(\boldsymbol{\beta}, \boldsymbol{\theta}) = -\frac{1}{2}\{n \log 2\pi + \log \det \mathbf{V} + (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\}. \quad (32.4.1)$$

For fixed  $\boldsymbol{\theta}$  only the last term involves  $\boldsymbol{\beta}$ , and it is the generalized least squares criterion of Definition 6.9.1, minimized by  $\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})$ . Writing

$$\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1}\mathbf{X}(\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X})^{-}\mathbf{X}^\top \mathbf{V}^{-1}, \quad (32.4.2)$$

the profile log-likelihood of Definition 7.3.1 is

$$\ell_P(\boldsymbol{\theta}) = -\frac{1}{2}\{n \log 2\pi + \log \det \mathbf{V} + \mathbf{y}^\top \mathbf{P}\mathbf{y}\}. \quad (32.4.3)$$

#### Lemma 32.4.1 — The residual operator $\mathbf{P}$

Let  $\mathbf{V}$  be positive definite and  $r = \text{rank}(\mathbf{X})$ . The matrix  $\mathbf{P}$  of [Eq. 32.4.2](#) does not depend on the choice of generalized inverse, and

- (a)  $\mathbf{P}\mathbf{X} = \mathbf{0}$ ,  $\mathbf{PVP} = \mathbf{P}$ ,  $\mathbf{P}$  is symmetric and  $\text{rank}(\mathbf{P}) = n - r$ ;
- (b)  $\mathbf{P}\mathbf{y} = \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$  and  $\mathbf{y}^\top \mathbf{P}\mathbf{y} = \min_{\mathbf{b}} (\mathbf{y} - \mathbf{X}\mathbf{b})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{b})$ ;
- (c) if  $\mathbf{V}$  depends smoothly on  $\boldsymbol{\theta}$  and  $\mathbf{V}_j = \partial \mathbf{V} / \partial \theta_j$ , then  $\partial \mathbf{P} / \partial \theta_j = -\mathbf{P}\mathbf{V}_j\mathbf{P}$ .

*Proof.* Write  $\mathbf{X}_* = \mathbf{V}^{-1/2}\mathbf{X}$  (Theorem 1.7.6). Then  $\mathbf{P} = \mathbf{V}^{-1/2}(\mathbf{I} - \mathbf{M}_*)\mathbf{V}^{-1/2}$ , where  $\mathbf{M}_* = \mathbf{X}_*(\mathbf{X}_*^\top \mathbf{X}_*)^{-}\mathbf{X}_*^\top$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X}_*)$ , which by Theorem 6.3.7 is the same for every generalized inverse and has rank  $r$ . Symmetry,  $\mathbf{P}\mathbf{X} = \mathbf{V}^{-1/2}(\mathbf{I} - \mathbf{M}_*)\mathbf{X}_* = \mathbf{0}$ ,  $\mathbf{PVP} = \mathbf{V}^{-1/2}(\mathbf{I} - \mathbf{M}_*)^2\mathbf{V}^{-1/2} = \mathbf{P}$  and the rank statement follow at once, which is (a). For (b),  $\mathbf{P}\mathbf{y} = \mathbf{V}^{-1}\mathbf{y} - \mathbf{V}^{-1}\mathbf{X}\hat{\boldsymbol{\beta}}$  by the definition of  $\hat{\boldsymbol{\beta}}$ , and  $\mathbf{y}^\top \mathbf{P}\mathbf{y} = \|(\mathbf{I} - \mathbf{M}_*)\mathbf{V}^{-1/2}\mathbf{y}\|^2$ , the squared distance from  $\mathbf{V}^{-1/2}\mathbf{y}$  to  $\mathcal{C}(\mathbf{X}_*)$ , which is the stated minimum. Part (c) is proved after [Theorem 32.4.2](#), which supplies a shorter route. ■

## 32.4.2 Restricted maximum likelihood

The trouble with Eq. 32.4.3 is visible in the simplest case. With  $\mathbf{V} = \sigma^2 \mathbf{I}$  it reduces to the profile of Theorem 7.3.1, whose maximizer is  $\hat{\sigma}^2 = \text{SSE}/n$ , biased downwards by the factor  $(n-p)/n$ . The likelihood has spent  $p$  dimensions fitting  $\boldsymbol{\beta}$  and then measured the spread of the data as if it had not. The repair is to compute the likelihood of a part of the data that never saw  $\boldsymbol{\beta}$ .

### Definition 32.4.1 — Error contrasts and restricted maximum likelihood

Let  $r = \text{rank}(\mathbf{X})$ . An **error contrast** is a random variable  $\mathbf{k}^\top \mathbf{Y}$  with  $\mathbf{X}^\top \mathbf{k} = \mathbf{0}$ ; its distribution does not involve  $\boldsymbol{\beta}$ . A **maximal set** of error contrasts is  $\mathbf{K}^\top \mathbf{Y}$  for an  $n \times (n-r)$  matrix  $\mathbf{K}$  of full column rank with  $\mathcal{C}(\mathbf{K}) = \mathcal{C}(\mathbf{X})^\perp$ . The **restricted** (or residual) **log-likelihood** is the log-likelihood of  $\mathbf{K}^\top \mathbf{Y}$ ,

$$\ell_R(\boldsymbol{\theta}) = -\frac{1}{2} \{ (n-r) \log 2\pi + \log \det(\mathbf{K}^\top \mathbf{V} \mathbf{K}) + \mathbf{y}^\top \mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{y} \}, \quad (32.4.4)$$

and the **REML estimate** of  $\boldsymbol{\theta}$  maximizes it over the parameter space.

### Theorem 32.4.2 — The restricted likelihood

In Definition 32.4.1, with  $\mathbf{V}$  positive definite:

- (a)  $\mathbf{K}^\top \mathbf{Y} \sim N_{n-r}(\mathbf{0}, \mathbf{K}^\top \mathbf{V} \mathbf{K})$  with  $\mathbf{K}^\top \mathbf{V} \mathbf{K}$  positive definite, so Eq. 32.4.4 is its log-likelihood.
- (b)  $\mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top = \mathbf{P}$  for every admissible  $\mathbf{K}$ . In particular the quadratic form in Eq. 32.4.4 equals  $\mathbf{y}^\top \mathbf{P} \mathbf{y}$  and does not depend on  $\mathbf{K}$ .
- (c) Two admissible  $\mathbf{K}_1, \mathbf{K}_2$  give restricted log-likelihoods differing by a constant free of  $\boldsymbol{\theta}$ . Hence the REML estimate does not depend on which maximal set of error contrasts is used.
- (d) If  $\text{rank}(\mathbf{X}) = p$ , then for any admissible  $\mathbf{K}$  with  $\mathbf{K}^\top \mathbf{K} = \mathbf{I}$ ,

$$\det(\mathbf{K}^\top \mathbf{V} \mathbf{K}) = \frac{\det \mathbf{V} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})}{\det(\mathbf{X}^\top \mathbf{X})}, \quad (32.4.5)$$

so that

$$\ell_R(\boldsymbol{\theta}) = \ell_P(\boldsymbol{\theta}) + \frac{p}{2} \log 2\pi - \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) + \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X}). \quad (32.4.6)$$

*Proof.*

- (a) By Theorem 3.3.1,  $\mathbf{K}^\top \mathbf{Y}$  is normal with mean  $\mathbf{K}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{0}$  and covariance  $\mathbf{K}^\top \mathbf{V} \mathbf{K}$ , which is positive definite because  $\mathbf{K}$  has full column rank and  $\mathbf{V}$  is positive definite. Its density is Theorem 3.2.2.

- (b) Fix
- $\mathbf{y}$
- and consider

$$\min\{\mathbf{z}^\top \mathbf{V}^{-1} \mathbf{z} : \mathbf{K}^\top \mathbf{z} = \mathbf{K}^\top \mathbf{y}\}.$$

The feasible set is  $\mathbf{y} + \mathcal{C}(\mathbf{X})$ : indeed  $\mathbf{K}^\top(\mathbf{z} - \mathbf{y}) = \mathbf{0}$  says  $\mathbf{z} - \mathbf{y} \in \mathcal{C}(\mathbf{K})^\perp = \mathcal{C}(\mathbf{X})$ . So the minimum equals  $\min_{\mathbf{b}} (\mathbf{y} - \mathbf{X}\mathbf{b})^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{y}^\top \mathbf{P} \mathbf{y}$  by Lemma 32.4.1(b). On the other hand the constrained problem can be solved directly. The criterion is strictly convex, so by Proposition 1.11.3 the minimizer satisfies  $\mathbf{V}^{-1} \mathbf{z} = \mathbf{K}\boldsymbol{\theta}$ , that is  $\mathbf{z} = \mathbf{V}\mathbf{K}\boldsymbol{\theta}$ ; the constraint gives  $\mathbf{K}^\top \mathbf{V}\mathbf{K}\boldsymbol{\theta} = \mathbf{K}^\top \mathbf{y}$ , and the minimum value is  $\boldsymbol{\theta}^\top \mathbf{K}^\top \mathbf{V}\mathbf{K}\boldsymbol{\theta} = \mathbf{y}^\top \mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{y}$ . The two expressions agree for every  $\mathbf{y}$ , and both are quadratic forms with symmetric matrices, so the matrices coincide.

- (c) Since  $\mathcal{C}(\mathbf{K}_1) = \mathcal{C}(\mathbf{K}_2)$  and both have  $n - r$  columns of full rank,  $\mathbf{K}_2 = \mathbf{K}_1 \mathbf{A}$  for a nonsingular  $(n-r) \times (n-r)$  matrix  $\mathbf{A}$ . Then  $\det(\mathbf{K}_2^\top \mathbf{V}\mathbf{K}_2) = (\det \mathbf{A})^2 \det(\mathbf{K}_1^\top \mathbf{V}\mathbf{K}_1)$ , while the quadratic forms are equal by (b). So the two log-likelihoods differ by  $-\log|\det \mathbf{A}|$ , which does not involve  $\boldsymbol{\theta}$ .
- (d) Let  $\mathbf{T} = [\mathbf{X}, \mathbf{K}]$ , which is nonsingular. Because  $\mathbf{X}^\top \mathbf{K} = \mathbf{0}$  and  $\mathbf{K}^\top \mathbf{K} = \mathbf{I}$ ,

$$\det(\mathbf{T}^\top \mathbf{T}) = \det \begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} = \det(\mathbf{X}^\top \mathbf{X}),$$

so  $(\det \mathbf{T})^2 = \det(\mathbf{X}^\top \mathbf{X})$  and  $\det(\mathbf{T}^\top \mathbf{V}\mathbf{T}) = \det(\mathbf{X}^\top \mathbf{X}) \det \mathbf{V}$ . Next,  $\mathbf{T}^{-1}$  has rows  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  stacked on  $\mathbf{K}^\top$ , as one checks by multiplying. Hence the leading  $p \times p$  block of  $(\mathbf{T}^\top \mathbf{V}\mathbf{T})^{-1} = \mathbf{T}^{-1} \mathbf{V}^{-1} \mathbf{T}^{-\top}$  is  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ . By Theorem 1.6.4 that block is the inverse of the Schur complement of the trailing block  $\mathbf{K}^\top \mathbf{V}\mathbf{K}$ , so by Theorem 1.6.3(b),

$$\det(\mathbf{T}^\top \mathbf{V}\mathbf{T}) = \det(\mathbf{K}^\top \mathbf{V}\mathbf{K}) \cdot \frac{\det(\mathbf{X}^\top \mathbf{X})^2}{\det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})}.$$

Equating the two expressions for  $\det(\mathbf{T}^\top \mathbf{V}\mathbf{T})$  gives Eq. 32.4.5, and substituting it together with (b) into Eq. 32.4.4 gives Eq. 32.4.6.

■

The proof of Lemma 32.4.1(c) is now immediate: differentiating  $\mathbf{P} = \mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top$  with  $\mathbf{K}$  fixed gives  $\partial \mathbf{P} / \partial \theta_j = -\mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{V}_j \mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top = -\mathbf{P} \mathbf{V}_j \mathbf{P}$ .

*Idea.* REML fits  $\boldsymbol{\theta}$  to the residual directions only. It is the general form of the familiar divisor  $n-p$ : with  $\mathbf{V} = \sigma^2 \mathbf{I}$ ,  $\text{rank}(\mathbf{X}) = p$  and an orthonormal  $\mathbf{K}$  (that is  $\mathbf{K}^\top \mathbf{K} = \mathbf{I}$ , as in Theorem 32.4.2(d)), Eq. 32.4.4 is the likelihood of  $\mathbf{K}^\top \mathbf{Y} \sim N_{n-p}(\mathbf{0}, \sigma^2 \mathbf{I})$ , maximized at  $\hat{\sigma}^2 = \|\mathbf{K}^\top \mathbf{y}\|^2 / (n-p) = \text{SSE} / (n-p) = s^2$ ; by Theorem 32.4.2(c) any other admissible  $\mathbf{K}$  gives the same answer. Ordinary maximum likelihood gives  $\text{SSE} / n$ . So  $s^2$ , the estimator every earlier chapter used, was the REML estimator all along. Its bias in a genuine mixed model is computed exactly in Proposition 32.4.3.

*Warning.* Since  $\mathbf{K}$  is determined by  $\mathcal{C}(\mathbf{X})$ , two models with *different fixed effects* have restricted likelihoods of different data, and comparing them is meaningless; Section 32.5 makes the rule explicit. Worse, Eq. 32.4.6 shows that the constant  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$  is part of the definition, and software often omits it; then the reported restricted log-likelihood changes if a column of  $\mathbf{X}$  is measured in thousands rather than units, by  $\log|\det \mathbf{S}|$  for the rescaling matrix  $\mathbf{S}$ . Comparisons of covariance structures with the *same*  $\mathbf{X}$  are unaffected, since the constant cancels.

### 32.4.3 Why maximum likelihood is biased, exactly

In the balanced one-way model both likelihoods can be maximized in closed form, and the bias can be computed rather than asserted.

#### Proposition 32.4.3 — Balanced one-way: ML and REML

In the balanced normal one-way random effects model, write  $\lambda_1 = \sigma^2 + m\sigma_a^2$  and  $\lambda_2 = \sigma^2$ , and treat  $(\lambda_1, \lambda_2)$  with  $\lambda_1 \geq \lambda_2 > 0$  as the parameters. Then, up to constants free of the parameters,

$$-2\ell_P = g \log \lambda_1 + (n - g) \log \lambda_2 + \frac{\text{SSB}}{\lambda_1} + \frac{\text{SSE}}{\lambda_2}, \quad (32.4.7)$$

$$-2\ell_R = (g - 1) \log \lambda_1 + (n - g) \log \lambda_2 + \frac{\text{SSB}}{\lambda_1} + \frac{\text{SSE}}{\lambda_2}. \quad (32.4.8)$$

The two criteria differ in one coefficient, but leave the interior of the parameter space at different moments.

- (a) When  $\text{MSB} \geq \text{MSE}$ , REML gives  $\hat{\lambda}_1 = \text{MSB}$ ,  $\hat{\lambda}_2 = \text{MSE}$ , so  $\hat{\sigma}^2 = \text{MSE}$  and  $\hat{\sigma}_a^2 = (\text{MSB} - \text{MSE})/m$ : the moment estimates of Proposition 32.1.1(c), which are unbiased.
- (b) When  $\text{MSB} \geq g \text{MSE}/(g - 1)$ , maximum likelihood gives  $\hat{\lambda}_1 = \text{SSB}/g$ ,  $\hat{\lambda}_2 = \text{MSE}$ , so  $\hat{\sigma}_{a,\text{ML}}^2 = \text{MSE}$  and

$$\hat{\sigma}_{a,\text{ML}}^2 = \frac{1}{m} \left( \frac{g-1}{g} \text{MSB} - \text{MSE} \right).$$

The right-hand side is the *unconstrained* maximizer whatever the data, it is non-negative exactly under the stated condition, and

$$\mathbb{E} \left\{ \frac{1}{m} \left( \frac{g-1}{g} \text{MSB} - \text{MSE} \right) \right\} - \sigma_a^2 = -\frac{\lambda_1}{gm}. \quad (32.4.9)$$

- (c) Otherwise the maximizer lies on the boundary  $\lambda_1 = \lambda_2$  and the method returns  $\hat{\sigma}_a^2 = 0$  with  $\hat{\sigma}^2$  the pooled mean square: for REML when  $\text{MSB} < \text{MSE}$ , for maximum likelihood on the wider range  $\text{MSB} < g \text{MSE}/(g - 1)$ . Between the two thresholds the REML estimate of  $\sigma_a^2$  is positive and the ML estimate is zero.

*Proof.* By Eq. 32.1.2,  $\mathbf{V} = \lambda_1 \mathbf{M} + \lambda_2 (\mathbf{I} - \mathbf{M})$ , so  $\det \mathbf{V} = \lambda_1^g \lambda_2^{n-g}$  and  $\mathbf{V}^{-1} = \lambda_1^{-1} \mathbf{M} + \lambda_2^{-1} (\mathbf{I} - \mathbf{M})$ . Since  $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$ ,  $\mathbf{V}^{-1} \mathbf{1} = \lambda_1^{-1} \mathbf{1}$ , so the generalized least squares estimate of  $\mu$  is  $\bar{y}$  and the residual is  $\mathbf{y} - \bar{y} \mathbf{1}$ . Splitting it by Theorem 6.5.1 into its parts in  $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$  and  $\mathcal{C}(\mathbf{Z})^\perp$ ,

$$\mathbf{y}^\top \mathbf{P} \mathbf{y} = (\mathbf{y} - \bar{y} \mathbf{1})^\top \mathbf{V}^{-1} (\mathbf{y} - \bar{y} \mathbf{1}) = \frac{\text{SSB}}{\lambda_1} + \frac{\text{SSE}}{\lambda_2}.$$

Substituting into Eq. 32.4.3 gives Eq. 32.4.7. For the restricted version,  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = n/\lambda_1$  and  $\mathbf{X}^\top \mathbf{X} = n$ , so the correction in Eq. 32.4.6 is  $-\frac{1}{2} \log(n/\lambda_1) + \frac{1}{2} \log n = \frac{1}{2} \log \lambda_1$ , which converts the coefficient  $g$  of  $\log \lambda_1$  into  $g - 1$  after multiplying by  $-2$ . That is Eq. 32.4.8.

Both right-hand sides have the form  $c_1 \log \lambda_1 + S_1/\lambda_1$  plus  $c_2 \log \lambda_2 + S_2/\lambda_2$ , a sum

of two functions of separate arguments, each strictly convex in  $1/\lambda$  and minimized at  $\lambda = S/c$ . So without the constraint  $\lambda_1 \geq \lambda_2$ , REML gives  $\hat{\lambda}_1 = \text{SSB}/(g-1) = \text{MSB}$  and  $\hat{\lambda}_2 = \text{SSE}/(n-g) = \text{MSE}$ , while maximum likelihood gives  $\hat{\lambda}_1 = \text{SSB}/g$  and the same  $\hat{\lambda}_2$ . These are feasible exactly when  $\hat{\lambda}_1 \geq \hat{\lambda}_2$ , that is when  $\text{MSB} \geq \text{MSE}$  for REML and when  $\text{SSB}/g = (g-1)\text{MSB}/g \geq \text{MSE}$ , i.e.  $\text{MSB} \geq g \text{MSE}/(g-1)$ , for maximum likelihood. Translating back through  $\sigma^2 = \lambda_2$  and  $\sigma_a^2 = (\lambda_1 - \lambda_2)/m$  gives (a) and (b), except for [Eq. 32.4.9](#). For that, use  $\text{E MSB} = \lambda_1$  and  $\text{E MSE} = \sigma^2$ :

$$\text{E} \left\{ \frac{1}{m} \left( \frac{g-1}{g} \text{MSB} - \text{MSE} \right) \right\} = \frac{1}{m} \left( \frac{g-1}{g} \lambda_1 - \sigma^2 \right) = \sigma_a^2 - \frac{\lambda_1}{gm}.$$

For (c), suppose the unconstrained minimizer of one of the two objectives violates  $\lambda_1 \geq \lambda_2$ , which by the previous paragraph happens exactly on the stated range. Each objective is a sum of two terms, one in each argument, each strictly unimodal on  $(0, \infty)$  and tending to  $+\infty$  at both ends, so a constrained minimum exists; were it interior to  $\{\lambda_1 > \lambda_2\}$  the gradient would vanish there, making it the infeasible unconstrained minimizer. Hence it lies on the boundary  $\lambda_1 = \lambda_2 = \lambda$ , where the objective becomes  $c \log \lambda + (\text{SSB} + \text{SSE})/\lambda$  with  $c = n$  for ML and  $c = n-1$  for REML, minimized at the corresponding pooled mean square. ■

The bias  $-\lambda_1/(gm)$  is  $\lambda_1$  divided by the total sample size: the price of the single degree of freedom used up by the grand mean, charged entirely to the between-group component. In the proficiency study,  $\lambda_1 = \sigma^2 + m\sigma_a^2 = 0.25^2 + 4(0.20)^2 = 0.2225$  and  $gm = 48$ , so the bias is  $-0.0046$ , which is 11.6 per cent of  $\sigma_a^2$ . A simulation of 40 000 studies reproduces it to the last recorded digit. Both the formula [Eq. 32.4.9](#) and that simulation describe the *interior* expression of [Proposition 32.4.3\(b\)](#); truncating it at zero, as the estimator itself does, can only raise the estimate, so the estimator's own bias is slightly smaller in modulus. Here the expression is negative in 2.0 per cent of the simulated studies and the truncated bias is  $-0.0045$ . With  $p$  fixed effects instead of one the same calculation gives a bias of order  $p/(gm)$ , which is why REML matters most when the fixed-effects model is large relative to the data.

## 32.4.4 Score equations and algorithms

### Proposition 32.4.4 — Score equations

Let  $\mathbf{V}_j = \partial \mathbf{V} / \partial \theta_j$  and  $\mathbf{P}$  be as in [Eq. 32.4.2](#). Then

$$\begin{aligned} \frac{\partial \ell_P}{\partial \theta_j} &= -\frac{1}{2} \{ \text{tr}(\mathbf{V}^{-1} \mathbf{V}_j) - \mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{y} \}, \\ \frac{\partial \ell_R}{\partial \theta_j} &= -\frac{1}{2} \{ \text{tr}(\mathbf{P} \mathbf{V}_j) - \mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{y} \}, \end{aligned} \tag{32.4.10}$$

The restricted score exceeds the profile score by  $\frac{1}{2} \text{tr} \{ (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{V}_j \mathbf{V}^{-1} \mathbf{X} \}$ , which is nonnegative whenever  $\mathbf{V}_j$  is nonnegative definite, as it is for a variance-component parameterization  $\mathbf{V} = \sum_j \theta_j \mathbf{Z}_j \mathbf{Z}_j^\top$ .

*Proof.* For the profile,  $\partial \log \det \mathbf{V} / \partial \theta_j = \text{tr}(\mathbf{V}^{-1} \mathbf{V}_j)$  and  $\partial(\mathbf{y}^\top \mathbf{P} \mathbf{y}) / \partial \theta_j = -\mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{y}$  by Lemma 32.4.1(c). For the restricted version, write  $\ell_R$  through Eq. 32.4.4 with a fixed  $\mathbf{K}$ : then  $\partial \log \det(\mathbf{K}^\top \mathbf{V} \mathbf{K}) / \partial \theta_j = \text{tr}\{(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{V}_j \mathbf{K}\} = \text{tr}(\mathbf{P} \mathbf{V}_j)$  by Theorem 32.4.2(b) and the cyclic property of the trace (Theorem 1.6.1), and the quadratic term differentiates as before. Subtracting, and using Eq. 32.4.2 for  $\mathbf{P}$ , gives the stated difference. Its value is the same for every generalized inverse, because  $\mathcal{C}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{V}_j \mathbf{V}^{-1} \mathbf{X}) \subset \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})$ , so take the Moore–Penrose inverse, which is nonnegative definite; when  $\mathbf{V}_j$  is nonnegative definite too, the difference is a trace of a product of two nonnegative definite matrices, hence nonnegative. ■

Setting Eq. 32.4.10 to zero gives equations that are not solvable in closed form outside balanced designs, and three families of algorithms are used.

**Fisher scoring.** The expected information for the restricted likelihood has entries  $\frac{1}{2} \text{tr}(\mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{V}_k)$ , so a Newton-type step costs one trace per pair of components. Convergence is fast near the maximum and unreliable far from it, and nothing keeps the iterates inside the parameter space.

**Average information REML.** Gilmour, Thompson and Cullis (1995) noticed that the average of the observed and expected information has entries  $\frac{1}{2} \mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{V}_k \mathbf{P} \mathbf{y}$ , which need no traces at all. Average information REML is the workhorse of large animal-breeding and genomic analyses.

**EM.** Treat  $\mathbf{u}$  as missing data (Dempster, Laird and Rubin, 1977). With  $\mathbf{R} = \sigma^2 \mathbf{I}$  and  $\mathbf{G} = \sigma_u^2 \mathbf{I}_q$ , the complete-data maximum likelihood estimates are averages of squares, so the E-step and M-step give the maximum likelihood update

$$\sigma_u^{2,\text{new}} = \frac{1}{q} \{ \|\hat{\mathbf{u}}\|^2 + \text{tr} \mathbf{S} \},$$

$$\sigma^2, \text{new} = \frac{1}{n} \{ \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{Z} \hat{\mathbf{u}}\|^2 + \text{tr}(\mathbf{Z} \mathbf{S} \mathbf{Z}^\top) \},$$

because  $E(\mathbf{u}^\top \mathbf{u} \mid \mathbf{y}) = \|\hat{\mathbf{u}}\|^2 + \text{tr} \mathbf{S}$  and  $E(\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon} \mid \mathbf{y}) = \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{Z} \hat{\mathbf{u}}\|^2 + \text{tr}(\mathbf{Z} \mathbf{S} \mathbf{Z}^\top)$ , where

$$\mathbf{S} = \text{Cov}(\mathbf{u} \mid \mathbf{y}) = (\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z})^{-1} = \mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G}$$

is the inverse of the trailing block of  $\mathbf{C}$  in Eq. 32.3.5, the matrix the push-through identity Eq. 32.3.7 was applied to. It is *not* the trailing block of  $\mathbf{C}^{-1}$  in Theorem 32.3.2(c), which carries a third term because  $\boldsymbol{\beta}$  has been estimated; that matrix gives an iteration maximizing neither likelihood. Each step is cheap, increases the likelihood and never leaves the parameter space; the price is slow convergence. A restricted version runs the same argument on the error contrasts. Laird and Ware (1982) introduced this algorithm to longitudinal data analysis, and it remains the most reliable starting procedure.

*Warning.* One starting value is not enough. Fit the Grunfeld panel of Example (Grunfeld’s panel with a random firm effect) with a random slope as well as a random intercept, on the logarithmic scale for the variances that keeps the iterates positive: a search begun with a negligible slope variance crawls along the boundary and stops at the random-intercept fit, with restricted log-likelihood  $-1176.53$  against the maximum  $-1159.01$ , a gap of 17.52 reported as successful convergence. Here thirty random starts all reach the same maximum, which is some evidence that it is the only one; but such likelihoods can also be genuinely multimodal, increasingly so as the covariance structure grows (Searle, Casella and McCulloch, 1992).

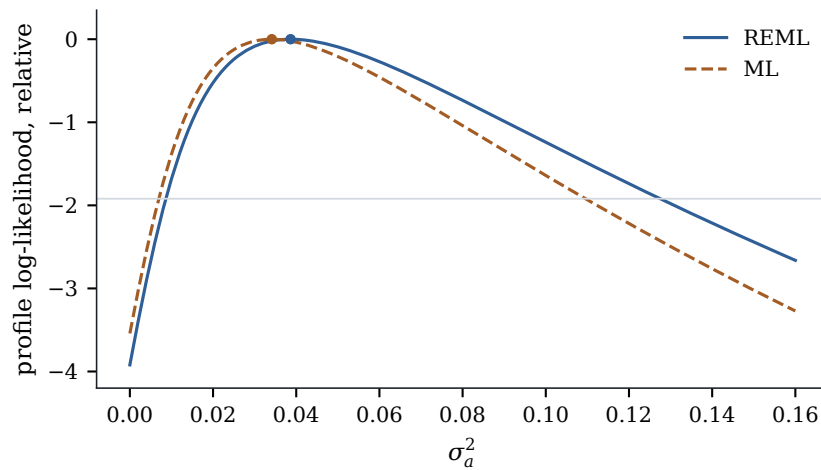


Figure 32.4.1: Profile log-likelihoods of  $\sigma_a^2$  for the proficiency study, with  $\sigma^2$  maximized out and each curve shifted to have maximum zero. The horizontal line is 1.92 below the maximum, the cut that gives an approximate 95% interval.

### 32.4.5 The proficiency study, computed

#### Example. (REML and ML for the twelve laboratories)

For the data of Example (A laboratory proficiency study), [Proposition 32.4.3](#) gives  $\hat{\sigma}^2 = 0.0620$  by both methods, and

$$\hat{\sigma}_{a,\text{REML}}^2 = 0.0386, \quad \hat{\sigma}_{a,\text{ML}}^2 = 0.0341,$$

the maximum likelihood estimate being 0.883 of the restricted one, because it replaces MSB by  $(g - 1)\text{MSB}/g$  before subtracting MSE. The restricted log-likelihood at its maximum is  $-8.228$  and the ordinary log-likelihood is  $-8.360$ ; the two are not comparable, being likelihoods of 47 and 48 numbers respectively. Both agree with the general matrix formulas coded from [Eq. 32.4.3](#) and [Eq. 32.4.4](#), and with statsmodels' `MixedLM`.

[Figure 32.4.1](#) plots the two profiles against  $\sigma_a^2$ , each with  $\sigma^2$  maximized out: the same shape in different places, the restricted one flatter on the right, which is why REML intervals are wider. Cutting the restricted profile 1.92 units below its maximum gives  $\sigma_a^2 \in (0.0086, 0.1275)$ , an interval comparable to the exact one of [Corollary 32.1.2](#) and available in models where no exact interval exists.

```
import numpy as np

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

X = np.ones((n, 1))
```

```

Z = np.repeat(np.eye(g), m, axis=0)

def loglik(s2, s2a, reml=True):
    """Log-likelihood (reml=False) or restricted log-likelihood of the variances."""
    V = s2a * (Z @ Z.T) + s2 * np.eye(n)
    L = np.linalg.cholesky(V)
    Vi_X = np.linalg.solve(V, X)
    W = X.T @ Vi_X #  $X' V^{-1} X$ 
    r = y - X @ np.linalg.solve(W, Vi_X.T @ y) # GLS residual
    quad = r @ np.linalg.solve(V, r) #  $y' P y$ 
    logdetV = 2 * np.sum(np.log(np.diag(L)))
    out = -0.5 * (n * np.log(2 * np.pi) + logdetV + quad)
    if reml:
        out += -0.5 * (np.linalg.slogdet(W)[1] - np.linalg.slogdet(X.T @ X)[1]
                        - X.shape[1] * np.log(2 * np.pi))
    return out

```

[Run this code online](#)

```

Y = y.reshape(g, m)
lab_mean = Y.mean(axis=1)
ms_between = m * np.sum((lab_mean - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - lab_mean[:, None]) ** 2) / (n - g)

reml_s2, reml_s2a = ms_within, (ms_between - ms_within) / m
ml_s2 = ms_within
ml_s2a = ((g - 1) / g * ms_between - ms_within) / m

print(f"REML  sigma^2 {reml_s2:.4f}  sigma_a^2 {reml_s2a:.4f}")
print(f"ML    sigma^2 {ml_s2:.4f}  sigma_a^2 {ml_s2a:.4f}")
print("restricted log-likelihood", round(loglik(reml_s2, reml_s2a), 4))
print("log-likelihood", round(loglik(ml_s2, ml_s2a, reml=False), 4))

```

[Run this code online](#)

Two hundred EM steps from a deliberately poor start reach those maximum likelihood values to nine decimal places, which is what a correct **S** buys; the trailing block of  $C^{-1}$  in its place converges instead to (0.0644, 0.0380), maximizing neither likelihood.

```

ZZt = Z @ Z.T
s2, s2a = 0.1, 0.1 # a deliberately poor starting value
for _ in range(200):
    V = s2a * ZZt + s2 * np.eye(n)
    Vi = np.linalg.inv(V)
    W = X.T @ Vi @ X
    beta_hat = np.linalg.solve(W, X.T @ Vi @ y)
    u_hat = s2a * Z.T @ Vi @ (y - X @ beta_hat)
    S = np.linalg.inv(np.eye(g) / s2a + Z.T @ Z / s2) # Cov(u | y), not  $(C^{-1})$ 
    resid = y - X @ beta_hat - Z @ u_hat
    s2a = (u_hat @ u_hat + np.trace(S)) / g
    s2 = (resid @ resid + np.trace(Z @ S @ Z.T)) / n

print(f"EM  sigma^2 {s2:.4f}  sigma_a^2 {s2a:.4f}")

```

[Run this code online](#)

## 32.4.6 Exercises

### 32.4.6.1 A. Check your understanding

#### Exercise A1.

Take  $\mathbf{V} = \sigma^2 \mathbf{I}$  and  $\text{rank}(\mathbf{X}) = p$ . Show from [Eq. 32.4.6](#) that  $-2\ell_R = (n - p) \log \sigma^2 + \text{SSE}/\sigma^2$  up to constants, and deduce that the REML estimate of  $\sigma^2$  is  $s^2$ . Then use Proposition 7.3.2 to find which multiple of SSE has the smallest mean squared error, and show that the maximum likelihood estimator beats  $s^2$  for some  $(n, p)$  and loses to it for others.

*Solution.* With  $\mathbf{V} = \sigma^2 \mathbf{I}$ :  $\log \det \mathbf{V} = n \log \sigma^2$ ,  $\mathbf{y}^\top \mathbf{P} \mathbf{y} = \text{SSE}/\sigma^2$  and  $\det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) = \sigma^{-2p} \det(\mathbf{X}^\top \mathbf{X})$ , so  $\log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) - \log \det(\mathbf{X}^\top \mathbf{X}) = -p \log \sigma^2$ ; adding this to  $-2\ell_P = n \log \sigma^2 + \text{SSE}/\sigma^2$  leaves  $(n - p) \log \sigma^2 + \text{SSE}/\sigma^2$ . Its minimizer is  $\text{SSE}/(n - p) = s^2$ . By Proposition 7.3.2 the best multiple of SSE is  $\text{SSE}/(n - p + 2)$ , which neither estimator in the question attains; writing  $k = n - p$ , the same proposition gives relative mean squared errors  $2/k$  for  $s^2$  and  $(2k + p^2)/n^2$  for  $\text{SSE}/n$ . Multiplying out  $n^2 = (k + p)^2$ , the inequality  $(2k + p^2)/n^2 \leq 2/k$  reduces to  $k(p - 4) \leq 2p$ . So maximum likelihood wins for every  $p \leq 4$ , whatever  $n$ , and loses once  $p$  is moderately large: with  $n = 100$  and  $p = 10$  it gives  $(180 + 100)/100^2 = 0.0280$  against  $2/90 = 0.0222$ . Fitting many fixed effects and then dividing by  $n$  is the bad case, which is the case REML exists for.

#### Exercise A2.

A colleague reports two restricted log-likelihoods for the same model fitted to the same data, differing by 6.91, the only difference between the runs being that one measured a regressor in units and the other in thousands. Explain the discrepancy using [Eq. 32.4.6](#), and confirm the value 6.91.

### 32.4.6.2 B. Practice

#### Exercise B1.

Verify [Theorem 32.4.2\(b\)](#) directly in the simplest case:  $n = 3$ ,  $\mathbf{X} = \mathbf{1}_3$ ,  $\mathbf{V} = \text{diag}(v_1, v_2, v_3)$ . Choose a basis  $\mathbf{K}$  of  $\mathbf{1}^\perp$ , compute  $\mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top$  and  $\mathbf{P}$ , and check they agree. Then change  $\mathbf{K}$  to a different basis and confirm that  $\det(\mathbf{K}^\top \mathbf{V} \mathbf{K})$  changes by a factor free of  $\mathbf{V}$ .

#### Exercise B2.

Two groups of sizes  $n_1, n_2$  have independent normal responses with a common mean and variances  $\sigma^2$  and  $c\sigma^2$  with  $c$  known. Write down  $\ell_P$  and  $\ell_R$  for  $\sigma^2$ , find both maximizers, and confirm that REML divides by  $n_1 + n_2 - 1$  and ML by  $n_1 + n_2$ .

#### Exercise B3.

Specialize the score equations [Eq. 32.4.10](#) of [Proposition 32.4.4](#) to the balanced one-way

model with  $\boldsymbol{\theta} = (\lambda_1, \lambda_2)$ , for which  $\mathbf{V} = \lambda_1 \mathbf{M} + \lambda_2 (\mathbf{I} - \mathbf{M})$ ,  $\mathbf{V}_1 = \mathbf{M}$  and  $\mathbf{V}_2 = \mathbf{I} - \mathbf{M}$ . Show that the restricted score equations are  $\hat{\lambda}_1 = \text{MSB}$  and  $\hat{\lambda}_2 = \text{MSE}$ , recovering [Proposition 32.4.3\(a\)](#) without any optimization.

*Solution.* Here  $\mathbf{P} = \lambda_1^{-1}(\mathbf{M} - \mathbf{P}_1) + \lambda_2^{-1}(\mathbf{I} - \mathbf{M})$ , so  $\mathbf{P}\mathbf{V}_1 = \lambda_1^{-1}(\mathbf{M} - \mathbf{P}_1)$  with trace  $(g-1)/\lambda_1$ , and  $\mathbf{y}^\top \mathbf{P}\mathbf{V}_1 \mathbf{P}\mathbf{y} = \text{SSB}/\lambda_1^2$ . The first equation in [Eq. 32.4.10](#) therefore reads  $(g-1)/\lambda_1 = \text{SSB}/\lambda_1^2$ , that is  $\lambda_1 = \text{SSB}/(g-1)$ . Likewise  $\text{tr}(\mathbf{P}\mathbf{V}_2) = (n-g)/\lambda_2$  and  $\mathbf{y}^\top \mathbf{P}\mathbf{V}_2 \mathbf{P}\mathbf{y} = \text{SSE}/\lambda_2^2$ , giving  $\lambda_2 = \text{SSE}/(n-g)$ . Using  $\mathbf{V}^{-1}$  instead of  $\mathbf{P}$  replaces  $g-1$  by  $\text{tr}(\mathbf{V}^{-1}\mathbf{V}_1)\lambda_1 = g$ , which is the ML answer.

### 32.4.6.3 C. Going deeper

#### Exercise C1.

Show that

$$\int \exp\{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})\} d\boldsymbol{\beta} = \exp\{\ell_R(\boldsymbol{\theta})\}$$

when  $\text{rank}(\mathbf{X}) = p$  and the constant  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$  is dropped from [Eq. 32.4.4](#). (Hint: complete the square in  $\boldsymbol{\beta}$  and use the normal integral.) This identity is the subject of [Section 32.6](#); state in words what it says about REML.

*Solution.* Write  $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{P}\mathbf{y} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$ , which is the decomposition behind [Lemma 32.4.1\(b\)](#). Integrating  $\exp\{-\frac{1}{2}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})\}$  over  $\mathbb{R}^p$  gives  $(2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}$ . Hence

$$\begin{aligned} \int e^\ell d\boldsymbol{\beta} &= e^{-\frac{1}{2}\{n \log 2\pi + \log \det \mathbf{V} + \mathbf{y}^\top \mathbf{P}\mathbf{y}\}} \\ &\quad \times (2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}, \end{aligned}$$

whose logarithm is [Eq. 32.4.6](#) without the  $\det(\mathbf{X}^\top \mathbf{X})$  term. In words: the restricted likelihood is the marginal likelihood of  $\boldsymbol{\theta}$  after integrating the fixed effects out with a flat prior.